

CH4203: Organic Synthesis Laboratory

Expt. No. 7: Organocatalytic Enantioselective Michael Reaction of Cyclohexanone with β -Nitrostyrene

Quiz

Marks: 5 (Each Question would carry 1.0 Mark)

(Q1) The mechanism of Organocatalytic Enantioselective Michael Reaction of Cyclohexanone with β -nitrostyrene in the presence of L-Proline follows:

- (a) Carbocation Pathway
- (b) Enamine Pathway
- (c) Iminium Pathway
- (d) Carbanion Pathway

Correct Answer: (b)

(Q2) NaBH₄ Reduction of Michael adduct of cyclohexanone and β -nitrostyrene would afford:

- (a) γ -hydroxy nitro derivative
- (b) β -hydroxy amine derivative
- (c) δ -hydroxy nitro derivative
- (d) δ -hydroxy amine derivative

Correct Answer: (c)

(Q3) The acid catalysed deprotection of *N*-Boc group of *N*-Boc L-prolinyl triazole derivative follows a:

- (a) S_N^{Ar}
- (b) S_N^I
- (c) S_N^2
- (d) S_N^i

Correct Answer: (b)

(Q4) A nitro reduction (under hydrogenation condition) followed by reductive amination of Michael adduct would result:

- (a) Azetidine derivative
- (b) Piperidine derivative
- (c) Piperazine derivative
- (d) Pyrrolidine derivative

Correct Answer: (d)

(Q5) NaBH₄ Reduction of Michael adduct (of cyclohexanone and β -nitrostyrene) followed by hydrogenation (Pd-C, 1 atm. of H₂ gas) would afford:

- (a) δ -hydroxy amine derivative
- (b) δ -hydroxy nitro derivative
- (c) γ -hydroxy amine derivative
- (d) γ -hydroxy nitro derivative

Correct Answer: (a)

(Q6) A Michael addition of thiophenol onto 2-cyclohexenone produces:

- (a) β -keto thioether
- (b) α -keto thioether
- (c) β -keto ether
- (d) γ -keto thioether

Correct Answer: (a)

(Q7) A partial reduction of an aliphatic nitro compound affords:

- (a) nitroso and hydroxylamine
- (b) amine and hydroxylamine
- (c) nitro and nitroso compound
- (d) nitro and amine compound

Correct Answer: (a)

(Q8) A Michael addition of thiophenol onto β -nitrostyrene produces:

- (a) β -nitro thioether
- (b) α -nitro thioether
- (c) δ -nitro thioether
- (d) γ -nitro thioether

Correct Answer: (a)

(Q9) Nitration of an aromatic ring using fuming HNO_3 could be considered as:

- (a) Aromatic nucleophilic substitution
- (b) Aromatic electrophilic substitution
- (c) Addition Elimination Reaction
- (d) Homolytic Aromatic Substitution

Correct Answer: (b)

(Q10) Out of the following statements which represents Mannich reaction:

- (a) aldehyde, amine, and ketone
- (b) aldehyde, nitroalkane, and ketone
- (c) acetylene, amine, and aldehyde
- (d) alkyl nitrile, amine, and ketone

Correct Answer: (a)